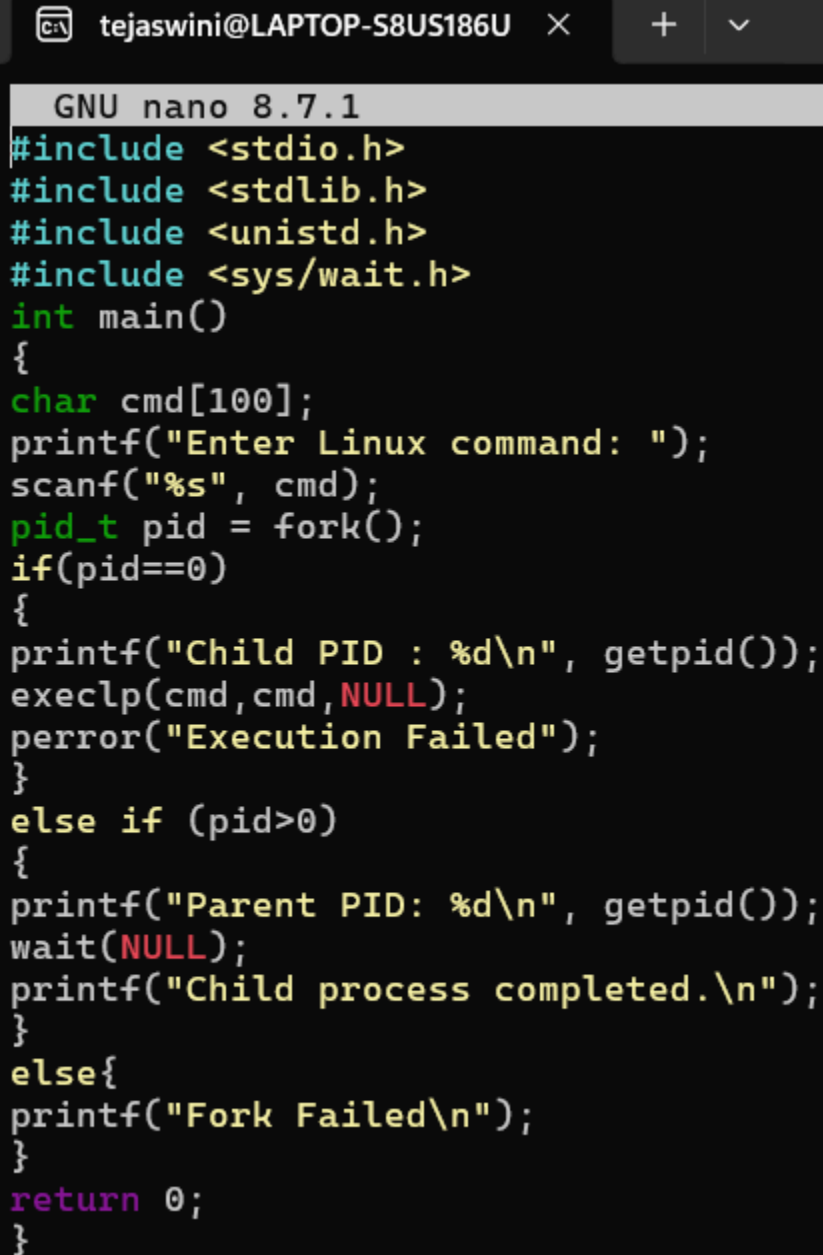


Practical 1 and 2:

Program 1:



The screenshot shows a terminal window with a dark background. The title bar at the top indicates the user is 'tejaswini@LAPTOP-S8US186U'. The terminal prompt is 'GNU nano 8.7.1'. The code is a C program that uses the fork() system call to create a child process. It prompts the user to enter a Linux command, which is then executed in the child process. The parent process waits for the child to complete and prints the child's PID. If the fork fails, it prints an error message. The code is as follows:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    char cmd[100];
    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid==0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd,cmd,NULL);
        perror("Execution Failed");
    }
    else if (pid>0)
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else{
        printf("Fork Failed\n");
    }
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/practical_1_2$ gcc ques1.c -o ques1
tejaswini@LAPTOP-S8US186U:~/OS_Lab/practical_1_2$ ./ques1
Enter Linux command: uname
Parent PID: 665
Child PID : 666
Linux
Child process completed.
```

Program 2:

```
GNU nano 8.7.1 ques_2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int src,dest;
    char buffer[1024];
    int bytes;
    src=open("input.txt",O_RDONLY);
    if(src<0)
    {
        printf("Cannot open source file\n");
        return 1;
    }
    dest=open("output.txt",O_WRONLY | O_CREAT | O_TRUNC, 0644);
    while((bytes = read(src,buffer,sizeof(buffer)))>0)
    {
        write(dest,buffer,bytes);
    }
    close(src);
    close(dest);
    printf("File copied successfully. \n");
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/practical_1_2$ gcc ques_2.c -o ques_2
tejaswini@LAPTOP-S8US186U:~/OS_Lab/practical_1_2$ ./ques_2
File copied successfully.
tejaswini@LAPTOP-S8US186U:~/OS_Lab/practical_1_2$
```